

Ex No: 2

Date:

A PYTHON PROGRAM TO IMPLEMENT SIMPLE LINEAR REGRESSION USING LEAST SQUARE METHOD

Aim:

To implement a python program for constructing a simple linear regression using least square method.

Algorithm:

Step 1: Import necessary libraries:

- pandas for data manipulation and matplotlib.pyplot for plotting.

Step 2: Read the dataset:

- Use the pandas `read\_csv` function to read the dataset (e.g., headbrain.csv).
- Store the dataset in a variable (e.g., `data`).

Step 3: Prepare the data:

- Extract the independent variable (X) and dependent variable (y) from the dataset.
- Reshape X and y to be 2D arrays if needed.

Step 4: Calculate the mean:

- Calculate the mean of X and

y. Step 5: Calculate the coefficients:

- Calculate the slope (m) using the formula:

$$m = \frac{\sum_{i=1}^n (X_i - \bar{X})(y_i - \bar{y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

- Calculate the intercept (b) using the formula: $b = \bar{y} - m\bar{X}$

Step 6: Make predictions:

- Use the calculated slope and intercept to make predictions for each X value:

$$\hat{y} = mx + b$$

Step 7: Plot the regression line:

- Plot the original data points (X, y) as a scatter plot.
- Plot the regression line (X, predicted_y) as a line plot.

Step 8: Calculate the R-squared value:

- Calculate the total sum of squares (TSS) using the formula: $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$
- Calculate the residual sum of squares (RSS) using the formula:

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Calculate the R-squared value using the formula: $R^2 = 1 - \frac{RSS}{TSS}$

Step 9: Display the results:

- Print the slope, intercept, and R-squared value.

Step 10: Complete the program:

- Combine all the steps into a Python program.
- Run the program to perform simple linear regression on the dataset.

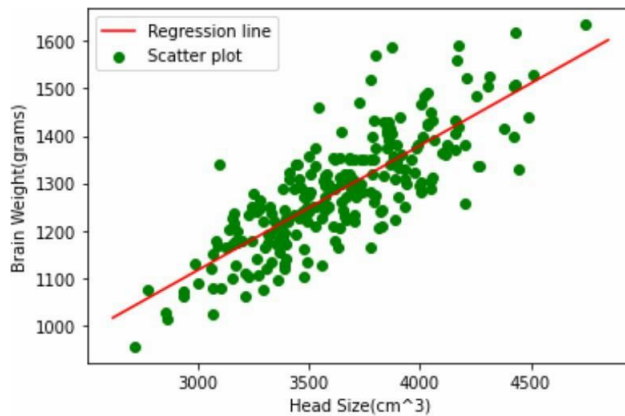
PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
data = pd.read_csv('headbrain.csv')
x, y = np.array(list(data['Head Size(cm^3)'])), np.array(list(data['Brain Weight(grams)']))
print(x[:5], y[:5])
[4512 3738 4261 3777 4177] [1530 1297 1335 1282 1590]
def get_line(x, y):
    x_m, y_m = np.mean(x), np.mean(y)
    print(x_m, y_m)
    x_d, y_d = x-x_m, y-y_m
    m = np.sum(x_d*y_d)/np.sum(x_d**2)
    c = y_m - (m*x_m)
    print(m, c)
```

```

return lambda x : m*x+c
lin = get_line(x, y)
X = np.linspace(np.min(x)-100, np.max(x)+100, 1000)
Y = np.array([lin(x) for x in X])
plt.plot(X, Y, color='red', label='Regression
line') plt.scatter(x, y, color='green', label='Scatter
plot') plt.xlabel('Head Size(cm^3)')
plt.ylabel('Brain
Weight(grams)') plt.legend()
plt.show()

```



```

def get_error(line_fuc, x, y):
    y_m = np.mean(y)
    y_pred = np.array([line_fuc(_) for _ in x])
    ss_t = np.sum((y-y_m)**2)
    ss_r = np.sum((y-y_pred)**2)
    return 1-(ss_r/ss_t)
get_error(lin, x, y)

```

In-built Package

```

from sklearn.linear_model import LinearRegression
x = x.reshape((len(x),1))

```

```
reg=LinearRegression()  
reg=reg.fit(x, y)  
print(reg.score(x, y))
```

RESULT:

Thus, the python program to implement simple linear regression using least square method for the given head brain dataset is analyzed and the linear regression line is constructed successfully